CIE AS PHYSICS (9702) • MARK SCHEME & MODEL ANSWERS

7.3 Doppler Effect for Sound Waves

Syllabus Section: 7. Waves **Paper Type:** Multiple Choice (Paper 1) **Total:** 14 Questions (14 Marks)

Question 1 Easy [1 mark]

Which of the following statements are true?

A When a source emitting a sound wave moves away from an observer, the observed frequency increases

B When a source emitting a sound wave moves away from an observer, the observed wavelength increases

C When a source emitting a sound wave moves towards an observer, the observed frequency decreases

D When a source emitting a sound wave moves towards an observer, the observed wavelength increases

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- When a source emitting a sound wave moves away from an observer, the waves are stretched
  - Therefore to an observer the wavelength increases

Option **A** is incorrect because

- When a source emitting a sound wave moves away from an observer, the observed frequency decreases as the waves are stretched

Option **C** and **D** is incorrect because

- When a source emitting a sound wave towards an observer, the waves are compressed
- Therefore the observed wavelength decreases and the frequency increases

Question 2 Easy [1 mark]

A source emits a sound wave of speed v and frequency f_s. The source moves towards an observer at a speed of v_s.

What is the observed frequency of the source?

A $\frac{f_s v}{v + v_s}$

B $\frac{f_s v_s}{v - v_s}$

C $\frac{f_s v}{v - v_s}$

D $\frac{f_s v_s}{v + v_s}$

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- The observed frequency f_o of a sound wave from a moving source is
  - $f_o = \frac{f_s v}{v \pm v_s}$
- When a source is moving towards an observer the denominator is $v - v_s$
- Therefore the observed frequency is
  - $f_o = \frac{f_s v}{v - v_s}$

Question 3 Easy [1 mark]

A train is moving towards a stationary observer. The train sounds its horn.

What are the correct changes in frequency, wavelength and pitch of the horn as heard by the observer?

	Frequency	Wavelength	Pitch
A.	higher	shorter	higher
B.	higher	shorter	lower
C.	lower	shorter	lower
D.	lower	longer	higher

A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- If the train is moving towards the observer then the wavelength of the sound waves is going to become shorter
  - This eliminates answer **D**
- If wavelength is shorter, then, because $c = f\lambda$, a decrease in wavelength will result in an increase in frequency
  - This eliminates answer **C**
- The pitch of a sound is linked to its frequency
  - A higher frequency means a higher pitch
  - Therefore the correct answer is **A**

Question 4 Easy [1 mark]

Which of the following equations is correct for the frequency of a source that is moving toward an observer?

A $f_s = \frac{f_0 v - v_s}{v}$

B $f_s = \frac{f_0 v + v_s}{v}$

C $f_s = \frac{f_o v}{v - v_s}$

D $f_s = \frac{f_o v}{v + v_s}$

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- The correct rearrangement of the doppler effect for sound equation to make the source frequency the subject is:
  - $f_0 = \frac{f_s v}{v \pm v_s} \rightarrow f_s = \frac{f_0 v \pm v_s}{v}$
- When the source is moving toward the observer the expression is $v - v_s$
- Therefore, the correct equation is:
  - $f_s = \frac{f_0 v - v_s}{v}$
- This is answer option **A**

Question 5 Easy [1 mark]

A car driving at a velocity of 10 m s^{-1} toward an observer sounds its horn with a frequency of 10 kHz. The speed of sound in air is 330 m s^{-1}.

What frequency does the observer hear?

A 8.9 kHz

B 9.7 kHz

C 10.1 kHz

D 10.3 kHz

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- The doppler effect equation for a source travelling toward an observer is $f_0 = \frac{f_s v}{v - v_s}$
- Where:
  - Frequency of source, $f_s = 10 \text{ kHz} = 10\,000 \text{ Hz}$
  - Velocity of source, $v_s = 10 \text{ m s}^{-1}$
  - Speed of sound in air, $v = 330 \text{ m s}^{-1}$
- Therefore, $f_0 = \frac{10\,000 \times 330}{330 - 10} = 10\,313 \text{ Hz}$
- 10 313 ≈ 10.3 kHz
  - This is answer **D**

Question 6 Medium [1 mark]

A police car siren emits a sound wave with a frequency f_s of 440 Hz. The car is travelling away from an observer at a speed of 30 m s^{-1}.

The speed of sound is 340 m s^{-1}.

Which of the following is the frequency the observer hears?

A 404 Hz

B 440 Hz

C 460 Hz

D 478 Hz

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- The Doppler effect is when the sound heard by an observer changes depending on the relative motion between the source and the observer
- The observed frequency of a sound wave from a moving source is
  - $f_o = \frac{f_s v}{v \pm v_s}$
- When the source is moving away from the observer, the waves spread out resulting in the wavelength increasing, so, the frequency decreases from the original emitted value.
- So the denominator is $v + v_s$
  - $f_o = \frac{f_s v}{v + v_s}$
- In this scenario, the source is the police car, and the person is the observer
- From the question
  - $f_s = 440$ Hz
  - «math xmlns=¨http://www.w3.org/1998/Math/MathML¨»«semantics»«mrow»«mi»v«/mi»«mo»=«/mo»«mn»340«/mn»«/mrow»«annotation encoding=¨application/vnd.wiris.mtweb-params+json¨»{¨fontFamily¨:¨Times New Roman¨,¨fontSize¨:¨18¨,¨autoformat¨:true,¨toolbar¨:¨«toolbar ref=`general`»«tab ref=`general`»«removeItem ref=`setColor`/»«removeItem ref=`bold`/»«removeItem ref=`italic`/»«removeItem ref=`autoItalic`/»«removeItem ref=`setUnicode`/»«removeItem ref=`mtext` /»«removeItem ref=`rtl`/»«removeItem ref=`forceLigature`/»«removeItem ref=`setFontFamily` /»«removeItem ref=`setFontSize`/»«/tab»«/toolbar»¨}«/annotation»«/semantics»«/math» m s^{-1}
  - «math xmlns=¨http://www.w3.org/1998/Math/MathML¨»«semantics»«mrow»«msub»«mi»v«/mi»«mi»s«/mi»«/msub»«mo»=«/mo»«mn»30«/mn»«/mrow»«annotation encoding=¨application/vnd.wiris.mtweb-params+json¨»{¨fontFamily¨:¨Times New Roman¨,¨fontSize¨:¨18¨,¨autoformat¨:true,¨toolbar¨:¨«toolbar ref=`general`»«tab ref=`general`»«removeItem ref=`setColor`/»«removeItem ref=`bold`/»«removeItem ref=`italic`/»«removeItem ref=`autoItalic`/»«removeItem ref=`setUnicode`/»«removeItem ref=`mtext` /»«removeItem ref=`rtl`/»«removeItem ref=`forceLigature`/»«removeItem ref=`setFontFamily` /»«removeItem ref=`setFontSize`/»«/tab»«/toolbar»¨}«/annotation»«/semantics»«/math» m s^{-1}
- Therefore
  - $f_o = \frac{440340}{340 + 30} = 404.32 \quad \text{Hz}$

Question 7 Medium [1 mark]

Figure 1.1 shows object *P*, moving at velocity v_s. It emits a sound wave of frequency f_s. The speed of sound is *v*.

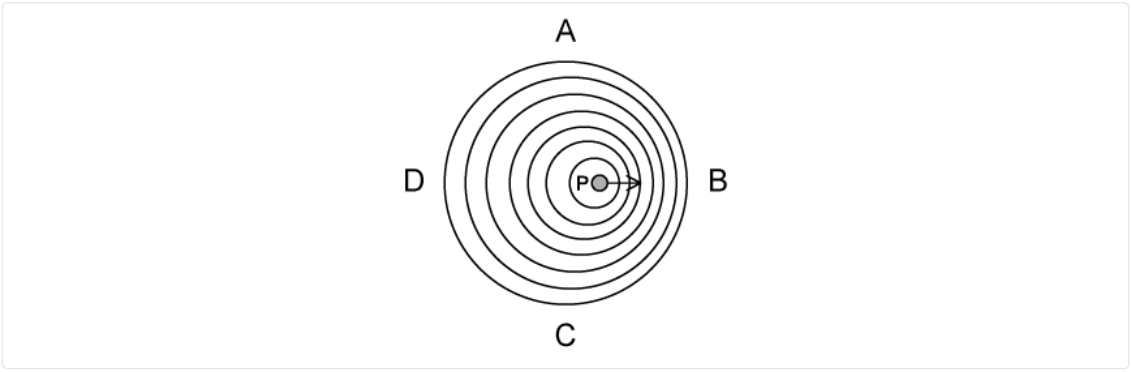


Fig. 1.1

At which point on the diagram will the observed frequency be $\frac{f_s v}{v + v_s}$?

A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- The observed frequency of a sound wave from a moving source is
  - $f_o = \frac{f_s v}{v \pm v_s}$
- When the source is moving away from the observer, the denominator is $v + v_s$
- At point **D**, the source is moving away, so option **D** must be correct

Option **A** and **C** are incorrect because:

- Point **A** and **C** are not directly in line with the moving source
- Therefore the relative velocity of the source to **A** and **C** is not v_s
  - The relative velocity will vary as *P* moves past **A** and **C**

Option **B** is incorrect because:

- When the source is moving towards the observer, the denominator is $v - v_s$
- The source is moving towards point **B,** therefore the observed frequency at point **B** is $\frac{f_s v}{v - v_s}$
- Make sure you understand when to use the + or − for this equation

Question 8 Medium [1 mark]

A source moving at constant velocity v_s, emits a sound wave of frequency 450 Hz. A stationary observer records a frequency of 300 Hz.

What is the value of v_s, and is it moving towards or away from the observer?

Speed of sound, v = 340 m s^{-1}

	v_s / m s^{-1}	Towards / Away
A.	170	Away
B.	170	Towards
C.	850	Away
D.	850	Towards

A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- The known quantities are:
  - Source frequency, f_s = 450 Hz
  - Observed frequency, f_o = 300 Hz
  - Speed of sound, v = 340 m s^{-1}
- The observed frequency is less than the source frequency
  - This means the wavelength has been stretched
  - The source must be moving **away** from the observer
- The observed frequency of a sound wave from a moving source is
  - $f_o = \frac{f_s v}{v \pm v_s}$
- When the source is moving away from the observer, the denominator is $v + v_s$
  - $f_o = \frac{f_s v}{v + v_s}$
- Rearrange for v_S
  - $f_o(v + v_s) = f_s v$
  - $v + v_s = \frac{f_s v}{f_o}$
  - $v_s = \frac{f_s v}{f_o} - v$
- Substitute the values into the equation
  - $v_s = \frac{450 \times 340}{300} - 340 = 170$ m s^{-1}
- This is option **A**

Question 9 Medium [1 mark]

A car of terrified observers try to get the car to start to drive away from a hungry T. rex, running toward them at a speed of 10 m s^{-1}. They hear the roar of the T. rex at a frequency of 170 Hz. The speed of sound in air is 340 m s^{-1}.

What is the frequency of the sound as emitted by the T. rex?

A 120 Hz

B 165 Hz

C 175 Hz

D 220 Hz

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- The Doppler equation for a source moving towards a stationary observer is $f_0 = \frac{f_s v}{v - v_s}$
- Rearranging to make the source frequency the subject gives $f_s = \frac{f_0 v - v_s}{v}$
- Where:
  - Observed frequency, $f_0 = 170$ Hz
  - Velocity of source, $v_s = 10$ m s^{-1}
  - Speed of sound in air, $v = 340$ m s^{-1}
- Therefore, $f_s = \frac{170 \times 340 - 10}{340} = 165$ Hz
  - This is answer **B**

A is incorrect as 100 m s^{-1} has been used for the speed rather than 10

C is incorrect as the operator + has been used rather than −

D is incorrect as 100 m s^{-1} has been used for the speed rather than 10 AND the operator + has been used rather than −

In many films where a T. rex chases a car, both dinosaur and car are moving. Luckily, you do not have to worry about relative motion when both source and observer are moving!

Question 10 Medium [1 mark]

A car travels in a straight line towards a stationary observer at a speed of 0.2v, where v is the speed of sound. The car's horn emits a pure sound of frequency f_s.

What is the frequency of the sound of the horn as heard by the observer, f_0?

A $f_0 = 0.83f_s$

B $f_0 = 1.00f_s$

C $f_0 = 1.20f_s$

D $f_0 = 1.25f_s$

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- The Doppler equation for a moving source is $f_0 = \frac{f_s v}{v \pm v_s}$
- As the car is moving **towards** the observer, the denominator becomes ($v - v_s$)
- Substituting 0.2v for v_s gives:
  - $f_0 = \frac{f_s v}{v - 0.2v} = \frac{f_s v}{0.8v} = f_s \frac{1}{0.8} = 1.25 f_s$

A is incorrect as 0.83 f_s is found when + is used in the denominator instead of −

B is incorrect as 0.2 has been used for v_s rather than v

C is incorrect as 1.20 f_s is found when $v + v_s$ is used incorrectly as the numerator

Question 11 Hard [1 mark]

Two runners, A and B, run past each other in opposite directions, both at speed u. A police car has its siren on. It is traveling at a speed of $10u$, in the same direction as Runner A. Runner A and Runner B observe the sound wave from the siren at frequencies of f_A and f_B respectively. The speed of sound is v.

What is the ratio of $\frac{f_A}{f_B}$?

A $\frac{v - 11u}{v + 9u}$

B $\frac{v}{v + 12u}$

C $\frac{v - 11u}{v - 9u}$

D $\frac{v}{v - 10u}$

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- The observed frequency of a sound wave from a moving source is
  - $f_0 = \frac{f_s v}{v \pm v_s}$
- Runner A is moving in the same direction as the police car
  - The police car is moving towards Runner A
- Therefore, the speed of the police car relative to Runner A is
  - $10\,u - u = 9\,u$
- The frequency of the sound wave that Runner A observes is
  - $f_A = \frac{f_s v}{v - 9u}$
- Runner B is moving in the opposite direction to Runner A
  - This means that Runner B is running towards the police car
- Therefore, the speed of the police car relative to Runner B is
  - $10\,u + u = 11\,u$
- The frequency of the sound wave that Runner B observes is
  - $f_B = \frac{f_s v}{v - 11u}$
- Therefore the ratio of f_A and f_B is
  - $\frac{f_A}{f_B} = \frac{f_s v}{v - 9u} \times \frac{v - 11u}{f_s v}$
  - $\frac{f_A}{f_B} = \frac{v - 11u}{v - 9u}$
- This is option **C**
- The velocity of the source must always be relative to the observer
  - If the question says 'relative to an observer' then this means the observer is stationary or the velocity of the observer has been taken into account
  - In this question, you needed to calculate the velocity of the source relative to the observer
- The police car moved towards both Runners A and B, even though they were running in opposite directions
  - Therefore the denominator for both scenarios was $v - v_s$

Question 12 Hard [1 mark]

Fig. 1.1 is a graph of the observed wavelength against time of a sound wave emitted from a single source. The emitted sound wave has a frequency f_s. The sound travels at a constant velocity directly towards an observer and a frequency of f_T is observed. It then travels directly away from the observer and a frequency of f_A is observed.

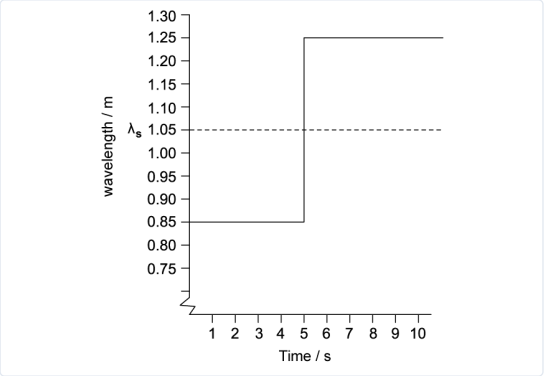


Fig. 1.1

What is the value of the ratio $\frac{f_A}{f_T}$?

A $\frac{25}{17}$

B $\frac{25}{21}$

C $\frac{17}{25}$

D $\frac{21}{25}$

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- The frequency of a wave is
  - $v = f\lambda$
  - $f = \frac{v}{\lambda}$
- When a source is moving towards an observer, the observed wavelength is less than the emitted wavelength
  - The graph shows that the wavelength is 0.85 m when the sound is moving toward the observer
- The observed frequency f_T when the source is moving towards the observer is
  - $f_T = \frac{v}{0.85}$
- When a source is moving away from an observer, the observed wavelength is greater than the emitted wavelength
  - The graph shows that the wavelength of the wave when it is moving away from the observer is 1.25 m
- The observed frequency f_A when the source is moving away from the observer is
  - $f_A = \frac{v}{1.25}$
- Therefore the ratio of f_A and f_T is
  - $\frac{f_A}{f_T} = \frac{v}{1.25} \times \frac{0.85}{v}$
  - $\frac{f_A}{f_T} = \frac{0.85}{1.25} = \frac{17}{25}$
- This is option **C**

Question 13 Hard [1 mark]

Fig 1.1 is a diagram of a source that has a mass of 500 g and is connected to a rope of length 20 cm. The rope is attached to point Q and has a tension of 800 N. The source travels around point Q anticlockwise in a circular path with a velocity of v_s. It emits a sound wave of frequency f_s. When the source is at point P, an observer stands directly in line to the direction of motion at point O.

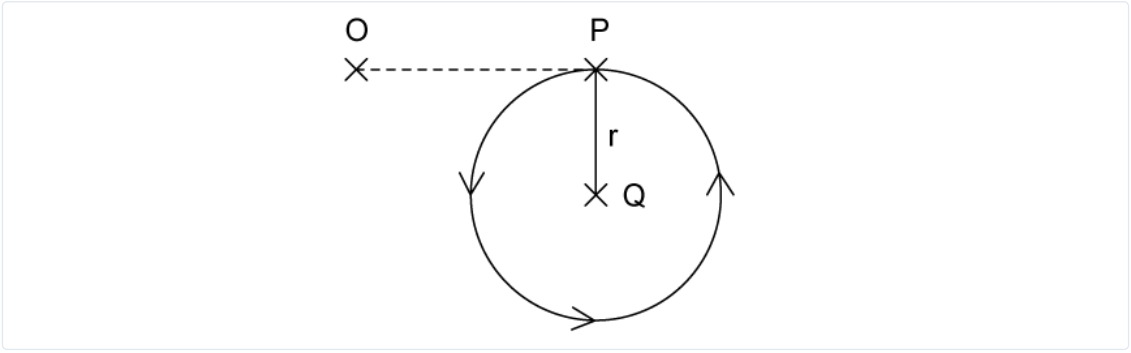


Fig 1.1

What is the value of v_s and $\frac{f_o}{f_s}$ when the source is at point P?

Speed of sound, v = 340 m s^{-1}

	v_s / m s^{-1}	$\frac{f_o}{f_s}$
A.	17.9	1.06
B.	17.9	0.950
C.	320	17.0
D.	320	0.515

A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- The known quantities are
  - Mass of the source, m = 0.5 kg
  - Length of string, r = 0.2 m
  - Tension of the string, F = 800 N
  - Speed of sound, v = 340 m s^{-1}
- The source is moving in a circular path, therefore
  - $F = \frac{mv_s^2}{r}$
  - $v_s^2 = \frac{Fr}{m}$
  - $v = \sqrt{\frac{Fr}{m}}$
- The centripetal force on the source is equal to the tension on the rope, so
  - $v_s = \sqrt{\frac{800 \times 0.2}{0.5}} = 17.9$ m s^{-1} (3 s.f)
- The observed frequency of a sound wave from a moving source is

- $f_0 = \frac{f_s v}{v \pm v_s}$

- At point **P** the source is moving directly towards the observer at point **O**
  - The denominator is therefore v - v_s
- The value of the ratio of f_0 and f_s is
  - $f_0 = \frac{f_s v}{v - v_s}$
  - $\frac{f_0}{f_s} = \frac{v}{v - v_s}$
  - $\frac{f_0}{f_s} = \frac{340}{340 - 17.9} = 1.06 (3 \text{ s.f})$
- This is option **A**
- Once you have calculated v_s, you know that the only two options available is **A** or **B**
- As the source is moving towards the observer, the observed frequency must be greater than the emitted frequency, therefore
  - $\frac{f_0}{f_s} > 1$
- By process of elimination, the answer must be option **A**

Question 14 Hard [1 mark]

A source moves directly in line with an observer at a constant velocity v_s and emits a sound wave with a wavelength of λ_s. The observed wavelength of the source is λ_0 and the ratio $\frac{\lambda_0}{\lambda_s}$ is equal to $\frac{5}{4}$.

What is the value of v_s?

Speed of sound, v = 340 m s^{-1}

A 68.0 m s^{-1}

B 85.0 m s^{-1}

C 612 m s^{-1}

D 765 m s^{-1}

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- The known quantities are
  - The ratio of λ_o and λ_s, $\frac{\lambda_o}{\lambda_s} = \frac{5}{4}$
  - Speed of sound, v = 340 m s^{-1}
- The wavelength of a wave is
  - $v = f\lambda$
  - $\lambda = \frac{v}{f}$
- The emitted wavelength is equal to
  - $\lambda_s = \frac{v}{f_s}$
- The observed wavelength is equal to
  - $\lambda_o = \frac{v}{f_o}$
- Therefore the ratio of λ_o and λ_s is
  - $\frac{\lambda_o}{\lambda_s} = \frac{v}{f_o} \times \frac{f_s}{v}$
  - $\frac{\lambda_o}{\lambda_s} = \frac{f_s}{f_o}$
- The observed frequency of a sound wave from a moving source is
  - $f_o = \frac{f_s v}{v \pm v_s}$
- If the source is moving away from an observer, the observed wavelength would be greater than the emitted wavelength so $\frac{\lambda_o}{\lambda_s} > 1$
  - As $\frac{\lambda_o}{\lambda_s}$ is equal to $\frac{5}{4}$ and this is greater than one, the source is moving away from the observer
- As the source is moving away, the denominator is therefore $v + v_s$
- The velocity of the source v_s is
  - $f_o = \frac{f_s v}{v + v_s}$
  - $v + v_s = \frac{f_s v}{f_o}$
  - $v_s = \frac{f_s v}{f_o} - v$
  - $v_s = \frac{5}{4} \times 340 - 340 = 85.0$ m s^{-1}
- This is option **B**